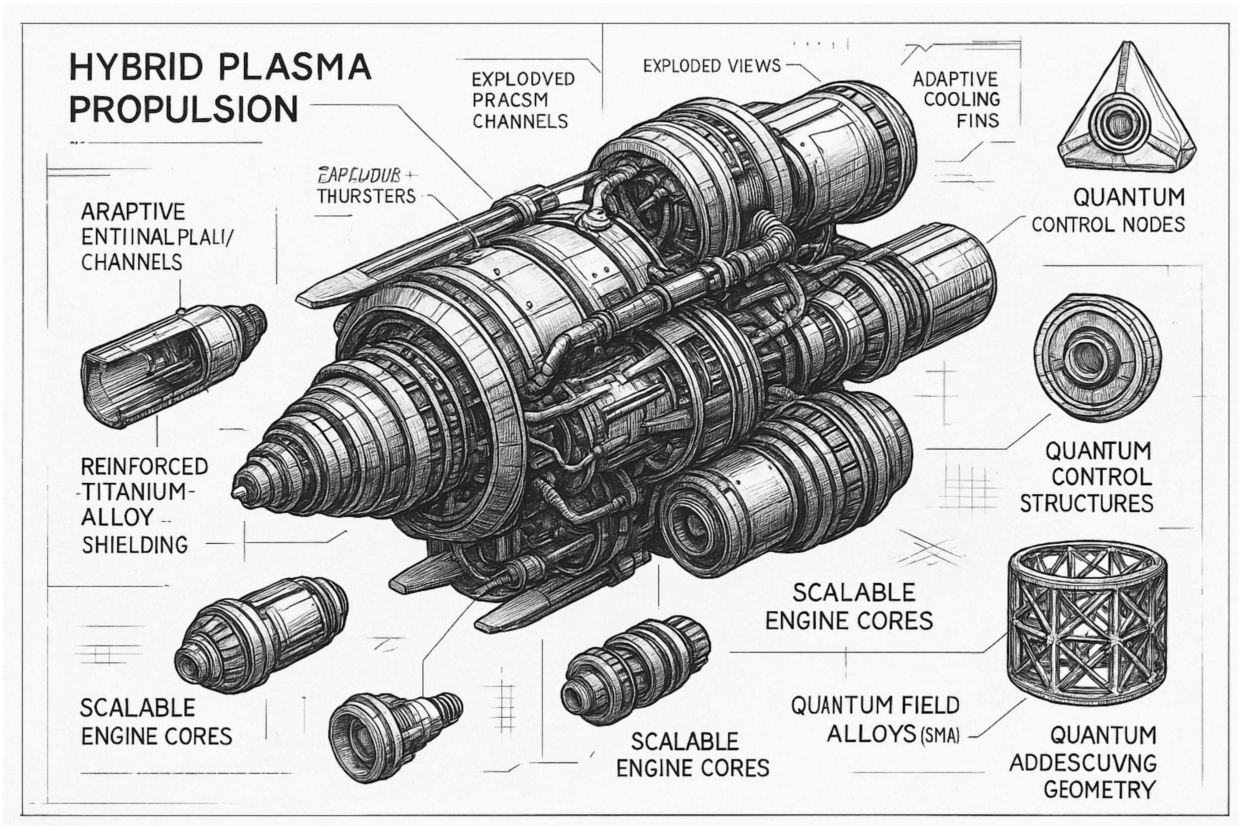
PROJECT SUMMARY: SOLARPARKERPROBE_MATERIALS(II)



JET ENGINE COMPONENTS

1. Diffuser - Slows down airflow and increases pressure.
2. Compressor - Increases air pressure via energy transfer from turbine.
3. Combustion Chamber - Location of air-fuel combustion.
4. Turbine - Converts thermal energy to mechanical energy.
5. Nozzle - Accelerates exhaust gases for thrust.

SPACECRAFT COMPONENTS

1. Jet Engines - Provide thrust.
2. Fuel System - Stores/supplies fuel.
3. Control System - Regulates engines/operations.

PLASMA & BLACK HOLE METRICS

1. Frolov Metrics - Spacetime near black holes.
2. Schwarzschild - Spherical black holes.
3. Kerr - Rotating black holes.

TURBINE & NOZZLE DESIGN

1. Turbine CAD Design - Precision modeling in CAD.

2. Nozzle CAD Design - CAD-based nozzle optimization.

PROPULSION MATERIALS
1. Fuel & Oxidizers   - Generate thrust.
2. Plasma Propulsion - Plasma-based systems.

BLACK HOLE RESEARCH
1. Spacetime Symmetries
2. Plasma Tensor Converter - Analyzes plasma-black hole interactions.

MATERIALS & SYSTEMS: SOLAR PARKER PROBE (SPP)

• Launch Mass: 685 kg
• Dimensions:
  - Height: 3 m
  - TPS Diameter: 2.3 m
  - Bus Diameter: 1 m
• C–C Thermal Protection System
• Bus: Hexagonal prism
• Solar Arrays:
  - Active, water-cooled
  - Output: ~364 W at closest approach
  - Area: 1.54 m²; Radiator: 4.0 m²
• Comms: 0.6 m HGA, 34W Ka-band, DL rate 163 kbps @ 1 AU

THERMAL CONTROL SYSTEM (SACS)

• Heat dissipation: 6400 W at 9.86 Rs
• Liquid Cooling Loop (Redundant Pump/Electronics)
• Temp Ranges:
  - Operative: +20°C to +150°C
  - Survival: +10°C to +190°C (wet), -80°C to -130°C (dry)
• Activation:
  - Prelaunch: Accumulator heated to 40–50°C
  - Radiators 2 & 3 active at 0.89 AU (Day 41)

Key Components:
• Accumulator (non-redundant)
• Pump Electronics (block redundant)
• Platens, Radiators, Valves
• Thermal Simulation & Testing (C1/C2 configs)

THERMAL TESTING

• C-1 Hot B6 (No TPS): Max 125°C, MLI fin in place
• C-2 Hot C6 (TPS at 300°C): Confirms TPS heat load
• C-1 Cold A2/B2: Validates minimum load at 1.02 AU
• Critical Cases: R23, Eclipse, Post-launch warm-up
• Data gathered to calibrate thermal model across mission phases

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CALCULATIONS: COMPRESSOR STAGES

Centrifugal Compressor:

INLET:
$C1 = 150$ m/s; $W1 = 250$ m/s; $U1 = 200$ m/s
OUTLET:
$C2 = 390$ m/s; $W2 = 150$ m/s; $U2 = 350$ m/s
$\beta 2$ = Flow angle; $\beta 2' = \beta 2 + 5°$

(a) Impeller Type:
Based on the velocity triangle, calculate $\beta 2$.
If $\beta 2 < 90° \rightarrow$ Forward-curved
$\beta 2 = 90° \rightarrow$ Radial
$\beta 2 > 90° \rightarrow$ Backward-curved

(b) Slip Factor (σ):
Use empirical models (Stodola, Wiesner) or:
$\sigma = 1 - (C\theta 2 / U2)$

(c) Stage Temperature Rise ($\Delta T0$):
$\Delta T0 = (U2 * C\theta 2 - U1 * C\theta 1) / cp$

Turboprop Dual Compressor System:

• Given:
- $T01 = 288$ K; $P01 = 101$ kPa
- $\pi 1 = 3$; $\sigma 1 = 0.94$; $\eta 1 = 0.86$; $\psi 1 = 1.04$
- $D1t = 0.5$ m; $D2 = 0.8$ m
- $\xi inlet = 0.65$; $M1rel = 1.0$

1. Mass Flow Rate ($\dot{m}$):
$\dot{m} = \rho * A * C1 = (P01 / RT01) * A * C1$

2. Inlet Velocity (C1):
$C1 = \sqrt{(2 * cp * \Delta T01)}$

3. Rotational Speed (N):
$U1 = \pi * D1 * N \rightarrow N = U1 / (\pi * D1)$

4. Compressor Power:
$P = \dot{m} * cp * \Delta T0$

AXIAL COMPRESSOR RELATIONS:

$\tan(\alpha 1) = [(1 - \Lambda) - \psi/2] / \varphi$
$\tan(\alpha 2) = [(1 - \Lambda) + \psi/2] / \varphi$
$\tan(\beta 1) = (\Lambda + \psi/2) / \varphi$
$\tan(\beta 2) = (\Lambda - \psi/2) / \varphi$

Temperature Rise Comparisons:

Axial:
$(C1/U)^2 = [(1 - \Lambda - \psi/2)^2 + \varphi^2]$
Centrifugal:
$\Delta T0\_axial / \Delta T0\_centrifugal = (Um/U2)^2$

If $Um = U2 \Rightarrow \Delta T0\_axial / \Delta T0\_centrifugal = rm / r2$

DEGREE OF REACTION (Λ):

$\Lambda = 1 - (\varphi^2 * A^2 * \sec^2\alpha 2 * \sec^2\alpha 1) / (AB * \tan\alpha 2 * \tan\alpha 1)$

APPLICATION: RADIATION-RESISTANT MATERIALS (SPP)

• Primary Shielding Material: Carbon–Carbon composite (C–C)
• Heat Shield (TPS): Ablative, layered for radiation protection
• Cooling Fluids: Water-based system with high latent heat
• High-temp MLI: Multi-Layer Insulation with aluminum/Mylar
• Key Design Constraint: Heat flux + radiation near perihelion (9.86 Rs)
• Redundant Sensors/Control logic for thermal monitoring

Shown are the CSPR temperatures as measured by the 32 temperature PT103 temperature sensors (eight per radiator) during the two activation events. The thermal “spike” at the beginning of each CSPR activation represents the orientation illustrated

The SACS was comprehensively calibrated over the operating temperature range (referenced at the pump CV) as a function of input power during the integrated thermal vacuum test (ITVT) in March 2017.
The SACS thermal model developed for the ITVT was correlated against the test data and was used and adjusted during the observatory thermal vacuum test in early 2018.
Prior to wedding CSPR2 and CSPR3, the SACS heat load based on the measured pump check valve (CV) temperature, 70°C, was estimated to be 1900 W using the correlated (flight) SACS thermal model.
Three days later, the umbra test was performed, where the heat load from the recently wetted CSPR2/CSPR3 was reduced to zero due to the umbra attitude, and the heat load to the SACS came exclusively through the SAs.

Correcting for the decrease in solar distance, the 1900 W predicted on 14 September 2018 (0.90 AU) became 2033 W (0.87 AU). After approximately 3 hours in umbra altitude, the measured steady-state temperature was ~33°C. This lined up very well with the model prediction for the 2033-W power input and four active radiators.
In flight, there is no way to independently or directly measure SACS input power, so it must be estimated based on temperature using the flight model

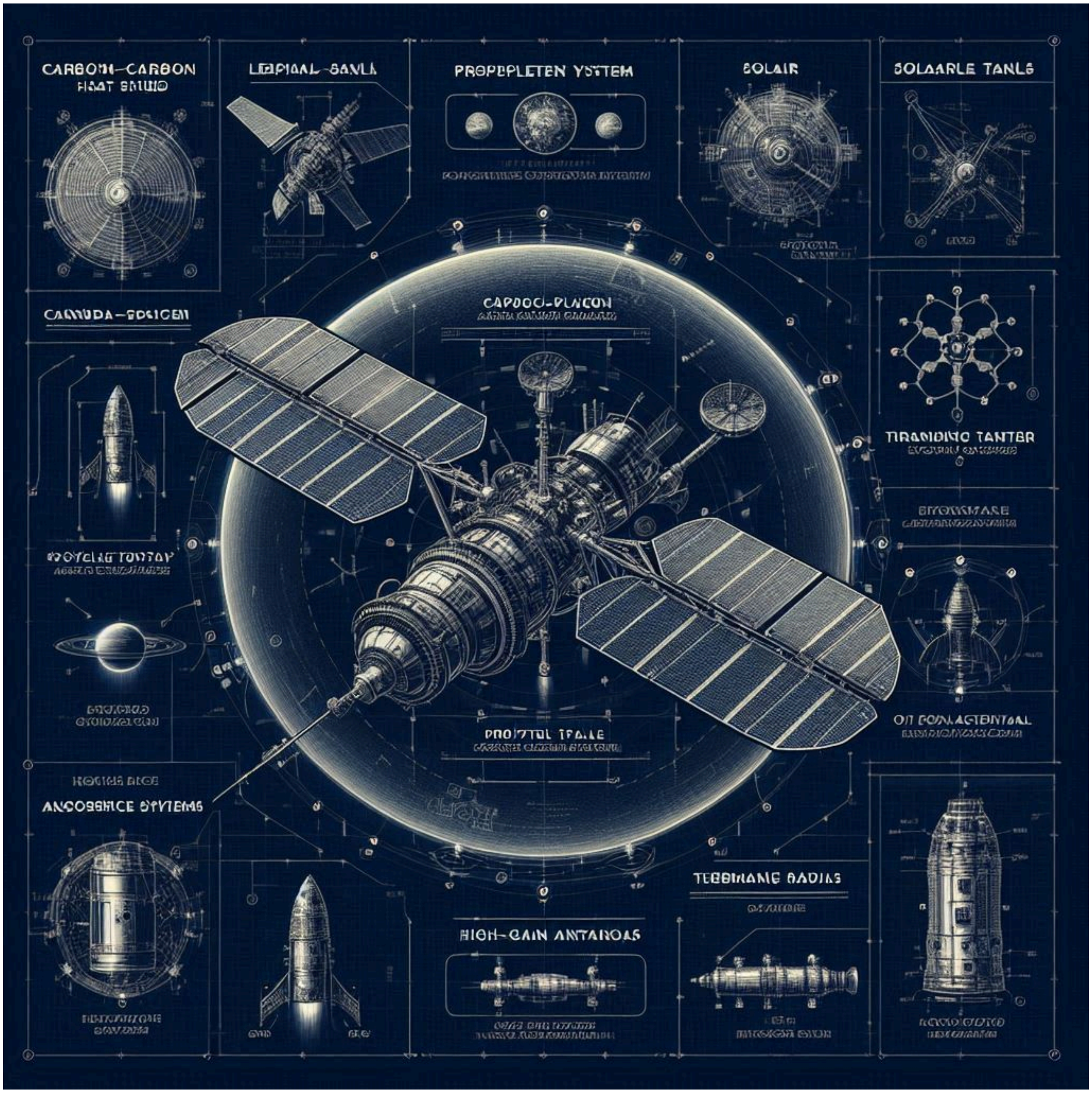
MATERIALS-DESIGN-PROMPT-METRICS-PLASMA-BLACK_HOLE_METRICS — Shown are the pump CV and accumulator temperatures (°C) as well as the system total pressure (psia) and delta-P (psid).

When inside of 0.70 AU (Umbra Attitude) the solar array flap angle is adjusted as a function of solar distance to keep the SACS operating temperature between 60 °C and 65 °C.
To date, the system has performed as designed.
26–30 August 2019, Newport News, VA — 30th Annual Thermal & Fluids Analysis Workshop

Spacecraft TCS Overview: The spacecraft (excluding the SACS) is a passive thermal design that utilizes louvers, multilayer insulation (MLI), and heaters along with waste electronics heat dissipation to maintain the core bus temperature between +10°C and +50°C.
The internally mounted SACS and propulsion components (the propellant tank and the accumulator, pressure transducers, latching valves, and fill and drain valves) are thermally coupled to the bus and rely on the bus bulk temperature to keep their respective temperatures above freezing (both hydrazine and water freeze at nearly the same temperature) during cold spacecraft conditions.
Two 20-blade and three 10-blade louvers are utilized to help minimize bus heater power and maintain the bulk bus temperature above 10°C.

The battery is thermally isolated from the bus and uses a seven-blade louver and software-controlled heaters to maintain its nominal temperature between 0°C and 15°C.
The majority of the high-powered radio frequency (RF) components are located on or near the −Z aft deck and utilize two of the 10-blade louvers to reduce bus heat leak when the RF system is powered off.

Instrument TCS Overview: The body-mounted instruments are thermally isolated from the bus and, with the exception of Solar Probe Analyzer (SPAN)-B, are located with the louvers on the “cold side” of the spacecraft (+X).

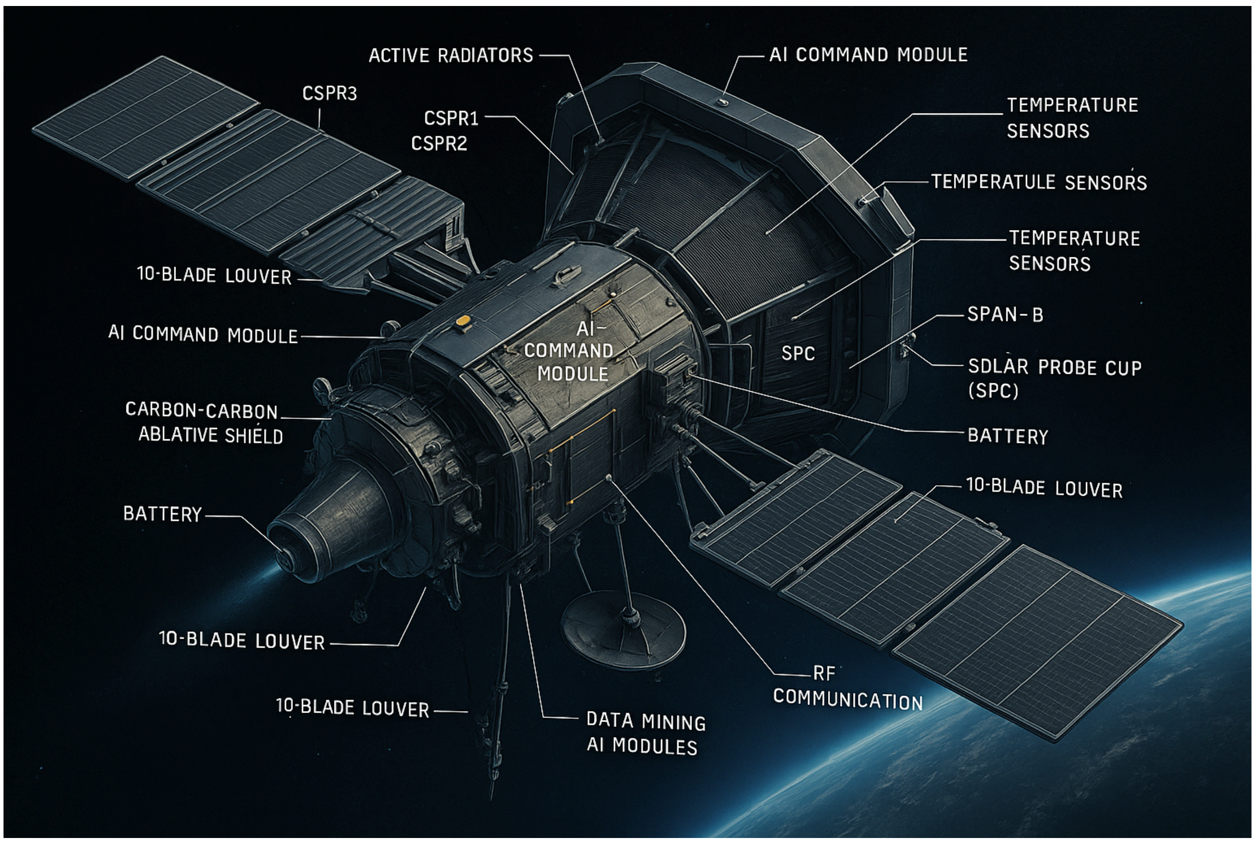


The truss structure assembly-mounted instruments, the four Electromagnetic Fields Investigation instrument suite (FIELDS) antennas, and the Solar Probe Cup (SPC) are completely isolated from the bus and are the only instruments designed to withstand the full solar constant when at minimum perihelion. The FIELDS magnetometers are located on a 3.4-m boom and also are thermally isolated from the bus.

During aphelion operation (outside of 0.70 AU), the spacecraft is tilted so as not to exceed 45° from +Z toward −X to allow for X-band (fan beam) and KA-band (HGA) communications.
Inside of 0.82 AU, this angle can be adjusted anywhere between umbra pointing and 45° to allow for proper HGA pointing toward Earth and also to relax the thermal input into the SACS, the externally mounted bus components, and the bus as long as the thermal attitude does not interfere with HGA-KA-band operations.

During this period, the externally mounted components on the −X side of the spacecraft experienced near-mission-maximum temperatures (0.70 AU/45° tilt), and some of the internally mounted components experienced mission-maximum temperatures as well.
These conditions are dependent on sun angle and will most likely repeat several more times as the spacecraft adjusts its attitude to allow proper HGA pointing for KA-band downlink.
So far in the mission, the spacecraft has experienced one of its two hot cases: during O-2 when near 0.70 AU and in aphelion-variable attitude while tilted 37° (1 March 2019) for KA-band downlink using the high-gain antenna (HGA).

In the second and more severe hot case, predicted for the mission minimum perihelion (9.86 RS), the external components will stay relatively cold and the temperature gradient inside of the spacecraft will be from top to bottom (+Z to −Z) instead of from side to side (−X to +X), so the internal components that were warm during this hot case will be a bit cooler during the perihelion hot case and vice versa.



1. General Structure (Macro-Architecture)
Shape: Cylindrical, elongated, stackable modules. External hexagonal cladding for interlock and heat dissipation.
Materials:

- Aluminum 6061-T6 (rings, supports)
- Carbon fiber (longitudinal reinforcements)
- Aluminized PET/Kapton with titanium coating (external cladding)

Modular Segmentation (Physical Layout)
Module 1 – Avionics Core / AI Node: Top/front. Titanium frame in aluminum ring for isolation. Jetson Nano, GPS, IMU, telemetry, diagnostics. Magnetic + bolted hatch with shielding.
Module 2 – Payload Bay: Quick-release magnetic latches + safety pins. DIN rail or 3D-printed supports. Passive/active cooling.
Module 3 – Power & Energy: LiPo 11.1V batteries in isolated bays. Heatsinks + radiator plate. Magnetic hot-swap connectors.

Module 4 – Propulsion System: Titanium/Inconel mount. Kevlar-wrapped composite tanks. Center fuel tube, injector, replaceable graphite nozzle.
Module 5 – Thermal & Magnetic Management: Vented aluminum rings. Brackets for coils or plasma devices.

Joining Mechanisms & Interfaces
Double titanium lock rings with rubber gaskets. Bolted flanges + centering pins. Magnetic connectors (polarity-coded), gold-plated pogo pins (EMI shielded). Central I2C/CAN bus, redundant power with self-healing fuses.

Exterior Shell / Thermal Armor
Structure: Carbon-fiber honeycomb (inner), airgel insulation (middle), PET/Kapton with TiO_2 (outer).
Style: Interlocking hex/rhomboid tiles with edge overlap. Turn-lock fasteners.
Color/Function: White/silver with blue/black vents. Heat-sensitive visual patches.

Attachment Points & External Hardware
Hardpoints at 45° axial/radial positions for instruments, antennas, etc. Service ports for refueling/data/battery, with retractable covers. CNC-machined lifting lugs.

Cooling Architecture (Passive/Hybrid)
Ends with titanium/aluminum gill-like vents, some retractable or SMA-based. Removable external radiators with internal liquid loop (ethylene glycol + nanoparticles).

an artificial black hole adapted for space travel. The main focus is to explain the plasma wave energy density required for radio emission generation in the plasmasphere of an exoplanet, whose density has been reduced.

Optical depth of induced conversion (τ_{sc}): This is related to the conversion of wave energy into radio emission. It is calculated with the formula:

$\tau_{sc} \approx -(\pi / 18\sqrt{3}) * (m_e \omega L v_{ph} / m_i c v_T) * w * L_n$

Radio flux at Earth (F): This is expressed with a formula involving various physical variables and parameters of the black hole, such as the background temperature (Tmi), distance (R_so), and Planck's constant (κB), among others:

$F = (3\kappa_B T_{mi} / c^2 m_e) * f_t^2 * (S_s / R_{so}^2) * \{\exp[\pi / 18\sqrt{3} * (m_e \omega L v_{ph} / m_i c v_T) * L_n * w - 1]\} * \exp(-\tau_{out}^c)$

Maser amplification condition: This is related to the generation of strong radio signals under certain conditions, where the optical depth of induced conversion ($|\tau_{sc}|$) is greater than the input optical depth (τ_{in}^c).

Plasma wave phase velocity (v_{ph}): This is linked to the plasma frequency (ω_p) and the wave number (k) through the relation:

$v_{ph} = \omega_p / k$

Generation at double plasma frequency (2ω): The behavior of the waves and their relationship with the double frequency, a relevant concept in the production of high-energy radio waves.

Differences in energy density: The image should show how the plasma wave energy density varies at different distances from the black hole, such as comparing the distances $R \lesssim 3R_p$ versus $R \gtrsim 25R_p$.

Centrifugal compressor:
Air enters through the center and is pushed outward (like in a washing machine centrifuge).

Used in small engines: turboprops, APUs, helicopters.

It has three parts:

- Rotor: spins and accelerates the air.
- Diffuser: slows the air and increases pressure.
- Manifold: sends compressed air to the combustion chamber.

Nuclear Physics and the Structure of Matter – Plasma-Based Accelerators

Plasma-based accelerators emerged from the innovative vision of John M. Dawson, who initially proposed accelerating particles at an angle to a traveling wave—an idea illustrated in Figures 10.50 and

10.51—so that the particles could gain speed faster than the phase velocity of the wave itself. This concept inspired a surge of experimental research, particularly in the United States, Japan, and England, aiming to develop a completely new class of particle accelerators. However, to date, only straightforward acceleration, where the angle is effectively zero, has been successfully implemented in practice.

Two primary concepts underpin early plasma accelerator designs: the **beatwave accelerator** and the **wakefield accelerator**. In the beatwave approach, two laser beams with frequencies and wave vectors (ω_0, k_0) and (ω_2, k_2) are superimposed to produce a beat frequency $\omega_1 = \omega_0 - \omega_2$, which is selected such that $\omega_1 \approx \omega_p$, the plasma frequency, which is significantly lower than the frequencies of the individual lasers. Correspondingly, their wave vector difference $k_1 = k_0 - k_2$ gives rise to a plasma wave with phase velocity $v_\varphi = \omega_1 / k_1 \approx \omega_p / k_p$. Since in a cold plasma the dispersion relation for plasma waves is nearly independent of wavelength, the wave number k_p can take the required value without affecting the wave properties. According to standard relations, the phase and group velocities of the light waves are governed by expressions like $v_{\varphi 0}^2 = \omega_0^2 / k_0^2 = c^2 + \omega_p^2 / k_0^2$, and the group velocity is then $v_{g0} = d\omega_0/dk_0 \approx c(1 + \omega_p^2 / \omega_0^2)^{(-1/2)}$, which also applies to the second laser line ω_2. In the limit where $\omega_p << \omega_0$, the group velocity of the light waves closely matches the phase velocity of the resulting plasma wave. This crucial match allows a laser pulse to continuously drive and accelerate particles trapped in the plasma wave over extended distances, maintaining synchronization between the pulse and the wave.

At higher plasma densities, the simplifying assumption $\omega_p << \omega_0$ becomes invalid, and the **critical density** n_c—where ω_p equals the laser frequency ω_0—becomes a defining parameter. It is given by the condition $n / n_c = \omega_p^2 / \omega_0^2$. Recasting the group velocity in terms of the critical density gives the relation $v_\varphi \approx v_{g0} = c(1 - n / n_c)^{(1/2)}$, which defines how light slows in plasma as the density approaches n_c. For example, n_c is about 1.0×10^{25} m^{-3} for the 10.6-μm CO_2 laser, 1.21×10^{25} m^{-3} for the 9.6-μm line, and 1.29×10^{25} m^{-3} for the 9.3-μm line. A beat frequency of 3×10^{12} s^{-1} between the 10.6-μm and 9.6-μm lasers corresponds to a plasma density of $n \approx 1.1 \times 10^{23}$ m^{-3}—an impressively high density that can be achieved either by pulsed plasma generation or directly by the laser beams themselves. Beat waves were first experimentally demonstrated at UCLA in 1985, and the first electron acceleration using beat waves was achieved there in 1993. Figure 10.52 illustrates the superposition of two oscillations differing in frequency by just 5%, producing a beat wave with a wavelength twenty times longer, in accordance with the beatwave formula $\omega_1 = \omega_0 - \omega_2$ and $k_1 = k_0 - k_2$.

In **wakefield acceleration**, instead of two lasers, a short and intense pulse of either photons or electrons is used to drive a plasma wave. The driving pulse travels at nearly the speed of light ($v \approx c$), generating a plasma wave with frequency f_p and wavelength $\lambda = c / f_p$. If this wave reaches a large enough amplitude, it can trap and accelerate electrons to relativistic velocities, effectively increasing their mass rather than their velocity due to the constraints of special relativity. An alternative method to achieve synchronization is to inject electrons at the correct phase of the wave by using photoemission from a solid surface, triggered by a pulsed laser precisely timed with the passing plasma wave, as depicted in Figure 10.53.

The relativistic factor γ for a plasma wave is a direct function of the phase velocity: $\gamma = [1 - (v_\varphi^2 / c^2)]^{(-1/2)}$. Using earlier definitions for n_c and ω_p, one obtains $\gamma = (n_c / n)^{(1/2)}$, which also implies $\gamma = \omega_0 / \omega_p$. This provides a concise relationship between the laser frequency, the plasma frequency, and the effective energy gain that particles can achieve while surfing the plasma wave.

Engine Designation	RD-120	RD-170	RD-253
Application (number of engines)	Zenit second stage (1)	Energia launch vehicle booster (4), Zenit first stage (1), Proton vehicle booster (1), and Atlas V (1)	Proton stage (1)
Oxidizer	Liquid oxygen	Liquid oxygen	N2O4
Fuel	Kerosene	Kerosene	UDMH
Number and types of turbopumps (TPs)	One main TP and two boost TPs	One main TP and two boost TPs	Single TP
Thrust control, %	Yes	Yes	±5
Mixture ratio control, %	±10	±7	±12
Throttling (full flow is 100%), %	85	40	None
Engine thrust (vacuum), kg	85,000	806,000	167,000
Engine thrust (SL), kg	—	740,000	150,000
Specific impulse (vacuum), sec	350	337	316
Specific impulse (SL), sec	—	309	285
Propellant flow, kg/sec	242.9	2393	528
Mixture ratio, O/F	2.6	2.63	2.67
Length, mm	3872	4000	2720
Diameter, mm	1954	3780	1500
Dry engine mass, kg	1125	9500	1080

Wet engine mass, kg	**1285**	**10,500**	**1260**

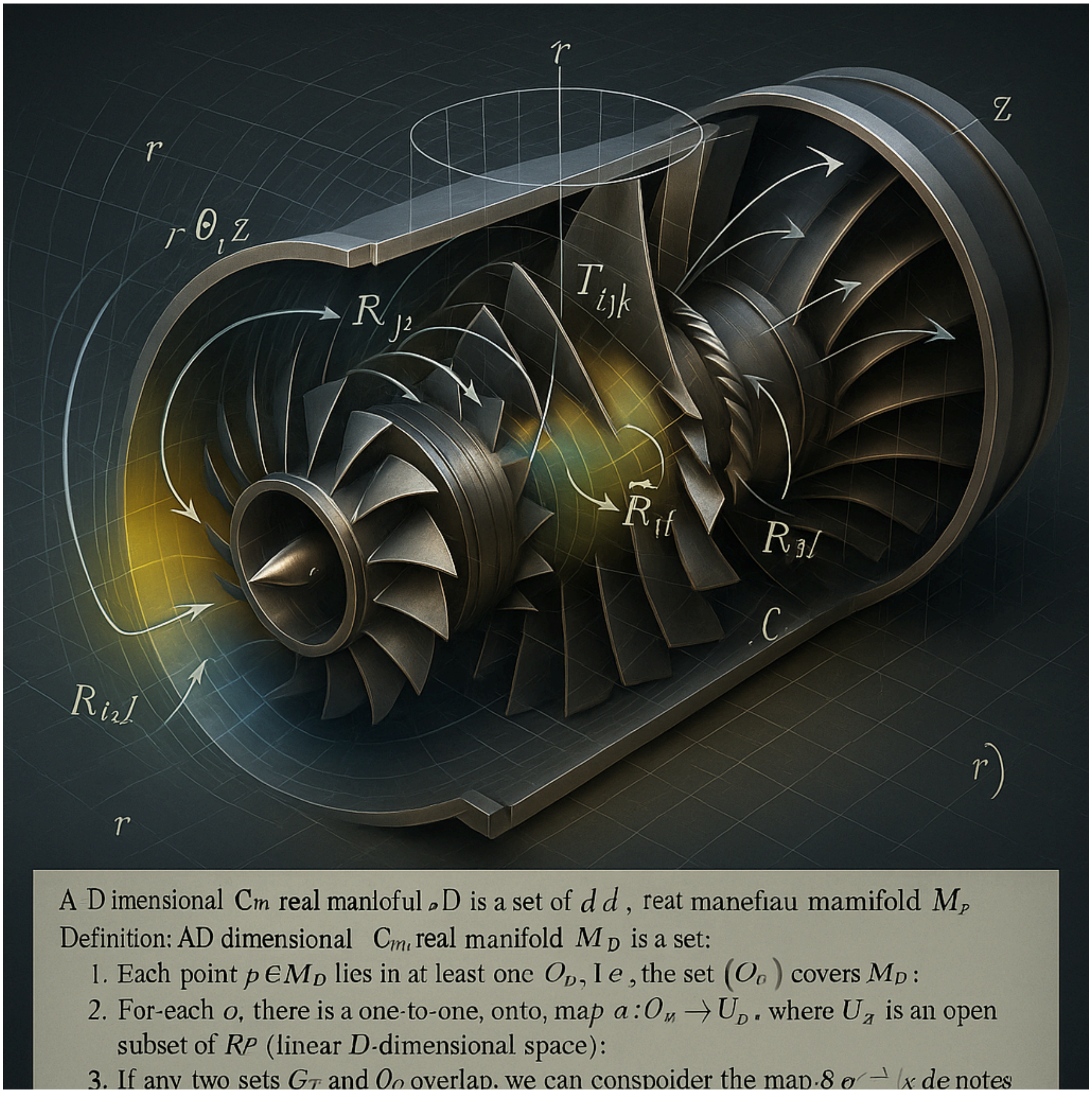
The concept revolves around designing a spacecraft propulsion system inspired by gravitational phenomena associated with black holes and neutron star collapses observed in various galaxies. The fundamental idea is to use the physics of gravity, particularly as described by black hole metrics, to create new propulsion mechanisms. The Penrose and Kruskal diagrams illustrate spacetime around a black hole formed from the collapse of scalar fields. The Penrose diagram shows observers at constant radii and their proper time, while the Kruskal diagram traces the paths of outgoing null rays emitted from the center of spacetime over time. These diagrams highlight key features such as real and apparent horizons, and the formation of spacelike singularities. Notably, the apparent horizon and event horizon differ because the spacetime in question is dynamic rather than static.

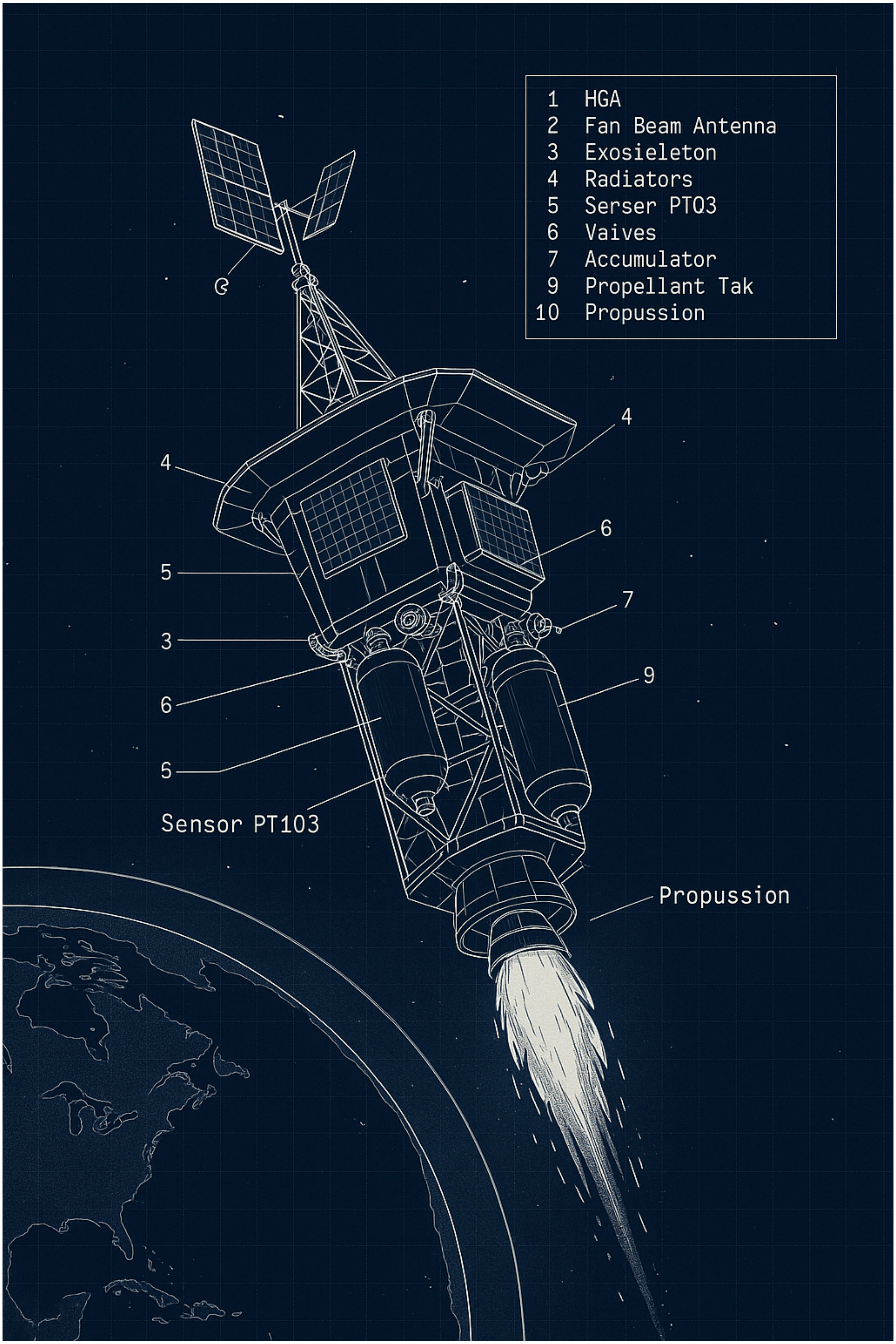
When considering a scalar field collapse characterized by an amplitude parameter p, two regimes arise: subcritical (p less than a critical threshold p*) where the field collapses and then reexpands, and supercritical (p greater than p*) where a black hole forms. Approaching the critical threshold from below leads to a spacetime structure resembling Minkowski space, while approaching from above leads to a Schwarzschild-like spacetime with zero mass. This creates two possible global spacetime structures with distinct features. The critical collapse spacetime is best studied using numerical simulations and characteristic coordinates, which follow light rays and enable the study of global structure and horizons.

The mathematical description involves spherically symmetric scalar field collapse in n+2 dimensions, with the metric expressed in terms of functions of coordinates and the curvature of the n-dimensional sphere. The reduced action formalism is used to derive equations of motion for the scalar field and metric functions. Variational principles yield evolution equations and constraint equations, which simplify under spherical symmetry. Numerical methods, including finite differencing on a null coordinate grid, solve these equations with second-order accuracy. The code iteratively finds field values at grid points to simulate the collapse evolution, capturing the formation of horizons and singularities.

The Penrose diagrams for subcritical and supercritical collapses show distinct behaviors: subcritical collapse exhibits oscillations of the scalar field mass with near-singular curvature at the center, while supercritical collapse reveals horizon formation and spacelike constant-radius lines inside the black hole. Gauge freedom in null coordinates allows adjusting the representation of spacetime so that the supercritical collapse diagram can be transformed to resemble the subcritical one, indicating deep underlying connections in the structure of spacetime near critical collapse thresholds. This suggests complex global behavior including naked singularities and potential Cauchy horizons requiring extra data for full description.

Overall, the material proposes adapting these advanced gravitational and spacetime collapse concepts into the design of spacecraft propulsion systems. By leveraging the behavior of scalar fields, horizons, and gravitational collapse phenomena, the propulsion design would harness exotic effects related to black holes, potentially enabling new modes of thrust and navigation through curved spacetime. Theoretical and numerical models provide a framework for simulating these conditions and informing engineering concepts for future spacecraft leveraging fundamental physics beyond classical rocket propulsion.




1 HGA
2 Fan Beam Antenna
3 Exosieleton
4 Radiators
5 Serser PTQ3
6 Vaives
7 Accumulator
9 Propellant Tak
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Sensor PT103
Propussion

Technical Internals of the Shuttle:

1. Thermal Control System (TCS):

CSPR radiators (8 per panel) with PT103 sensors.

SACS (Solar Array Cooling System): Thermally correlated model, with estimated power input (1900–2033 W).

Louvers (thermal vents) in solar shade for passive dissipation.

2. Power Unit:

Solar arrays (SAs) with adjustable flaps based on solar distance (between 0.70–0.90 AU).

Thermally insulated batteries, operating between 0°C and 15°C with dedicated software and louvers.

3. Propulsion and Fuel System:
Propellant tanks thermally coupled to the bus.

Check valves (CVs), fill and drain valves.

Pressure accumulators with transducers.

4. Bus Exostructure:
MLI (Multilayer Insulation).

Aphelion/umbra-optimized geometry.

Truss-like structure for antennas and instruments.

5. AI Core / Data Mining Center:
Dedicated module for real-time AI scientific data processing.

Includes radiation-tolerant, carbon-carbon-shielded quantum computing units.

6. Communication System:
X-band and KA-band antennas (HGA).

Dynamic orientation between 0°–45° based on solar distance to cool and maintain Earth link.

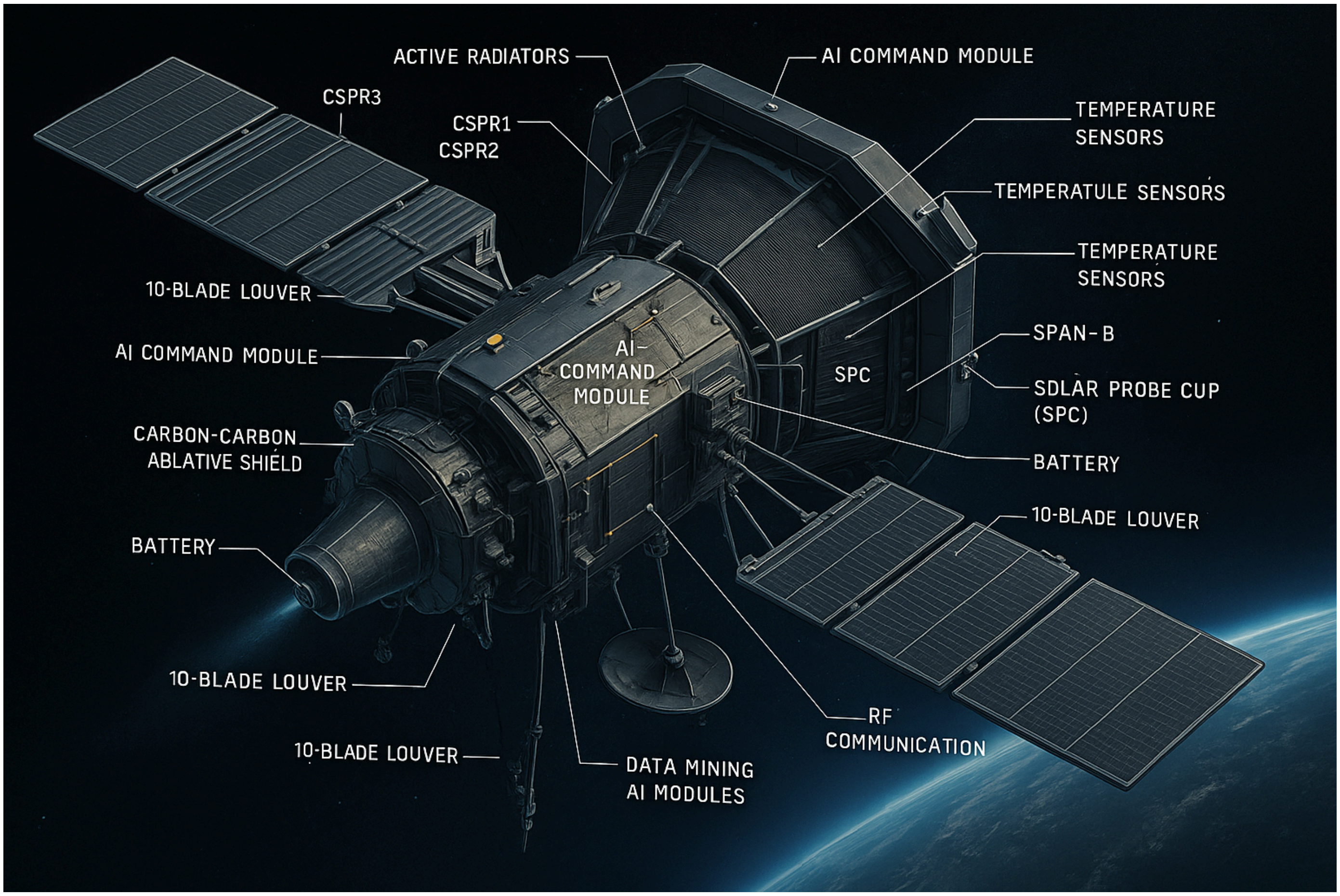
7. Scientific Instruments and Sensors:
FIELDS Suite (EM sensors).

Solar Probe Cup (SPC).

Magnetometers on a 3.4 m mast, thermally insulated.

8. Construction Materials:
Carbon-carbon fiber structure (high thermal resistance).

Internal components coated with radiation-resistant insulating materials (ceramic polymers).


ACTIVE RADIATORS
AI COMMAND MODULE
CSPR3
CSPR1
CSPR2
TEMPERATURE SENSORS
TEMPERATULE SENSORS
TEMPERATURE SENSORS
10-BLADE LOUVER
AI COMMAND MODULE
AI-COMMAND MODULE
SPAN-B
SPC
SDLAR PROBE CUP (SPC)
CARBON-CARBON ABLATIVE SHIELD
BATTERY
BATTERY
10-BLADE LOUVER
10-BLADE LOUVER
10-BLADE LOUVER
RF COMMUNICATION
DATA MINING AI MODULES